Monetchat AI Marketplace Deployment – Progress Summary

1. Infrastructure Setup

You are running the system on a **Windows Server with WSL + Docker Compose**.

The application stack deployed:

- **App:** Node.js / Next.js Monetchat application
- **Database:** PostgreSQL
- **Vector Database:** Qdrant
- **AI Models:** Ollama (running on Windows host)

Docker services currently running:

- Monetchat_app
- Monetchat_postgres
- Monetchat_qdrant

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2. Database Initialization

PostgreSQL was verified and configured.

Tables created successfully:

```
categories
chat_messages
chat_sessions
contact_logs
countries
product_images
products
prohibited_keywords
```

```
regions
search_logs
sellers
users
```

Database user:

```
Monetchatuser
```

Database name:

```
Monetchat
```

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3. Database Seeding

The system seed script was executed:

```
npm run db:seed
```

Seeded data includes:

- Countries
- Kuwait regions
- Product categories
- Prohibited keywords

This prepares the system for listings and moderation.

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4. Product Data Import

You imported marketplace listings from:

```
C:\inetpub\wwwroot\pickpickdata\q84sale_listings.json
```

After import the database contains:

```
576 products
```

Verified with:

```
select count(*) from products;
```

Result:

```
576
```

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5. Qdrant Vector Database

Qdrant service was created and configured.

Collection created:

```
products
```

Verification:

```
curl http://localhost:6333/collections
```

Result:

```
products
```


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6. Hybrid Search Pipeline

The application is designed to use:

```
Postgres → structured data
Qdrant → vector similarity
Ollama → embeddings + AI chat
```

Workflow:

```
Product text
     ↓
Embedding model
     ↓
Vector stored in Qdrant
     ↓
Hybrid ranking search
```

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7. Major Issue Encountered

During indexing we encountered:

```
ECONNREFUSED 127.0.0.1:11434
```

Meaning:

The Docker container was trying to connect to Ollama **inside the container**, but Ollama was running **on the Windows host**.

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8. Ollama Fix

We solved the networking issue by:

Step 1 – Reconfiguring Ollama

Changed Ollama to listen on all interfaces:

```
OLLAMA_HOST=0.0.0.0:11434
```

Verification:

```
netstat -ano | findstr 11434
```

Result:

```
0.0.0.0:11434 LISTENING
```

Step 2 – Determining the Windows Host IP

Detected WSL gateway:

```
192.168.192.1
```

This is now the **correct address for containers to reach Ollama**.

Step 3 – Updating Application Environment

.env was updated to replace:

```
127.0.0.1:11434
localhost:11434
host.docker.internal:11434
```

with

```
192.168.192.1:11434
```

Step 4 – Rebuilding the Stack

Docker environment rebuilt:

```
docker compose down
docker compose up -d --build
```

All containers restarted successfully.

9. Current System State

Infrastructure status:

Component	Status
App	Running
Postgres	Running
Qdrant	Running
Ollama	Running
Products	576 loaded
Qdrant collection	Created

10. Remaining Step

Final step in progress:

```
scripts/reindex-qdrant.ts
```

This will:

1. Generate embeddings using:

```
nomic-embed-text
```

2. Store vectors in:

```
Qdrant → products collection
```

3. Enable:

```
Semantic AI product search
```

Final Architecture


```
User Query
     ↓
AI Query Understanding (Ollama)
     ↓
Hybrid Search
 ┌─────────────┐
 │ Postgres    │
 │ metadata    │
 └─────────────┘
        +
 ┌─────────────┐
 │ Qdrant      │
 │ vectors     │
 └─────────────┘
        ↓
Ranked Results
        ↓
AI Response
```

Next Stage After Indexing

Once indexing finishes, the system will support:

AI Marketplace Search

Examples:

```
"cheap toyota corolla in kuwait"
"iphone under 100 kd"
"furniture near salmiya"
```

AI Chat Marketplace

Example:

```
User: find me a good used sofa under 40 KD
AI: suggests listings from Qdrant
```
